BORDER OPTICS

Maps & Plots

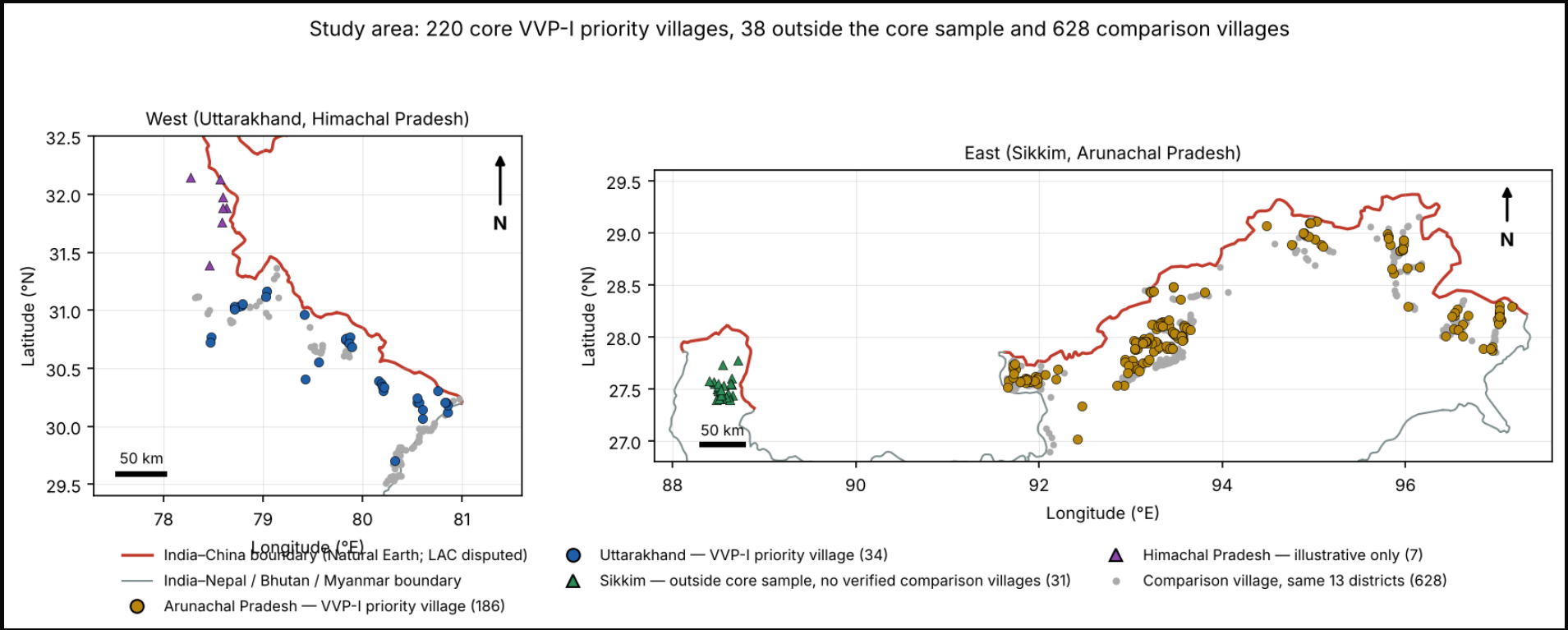
Evaluating a Place-Based Border Programme from Space:
Evidence and Indicator Limits from India's Vibrant Villages Programme

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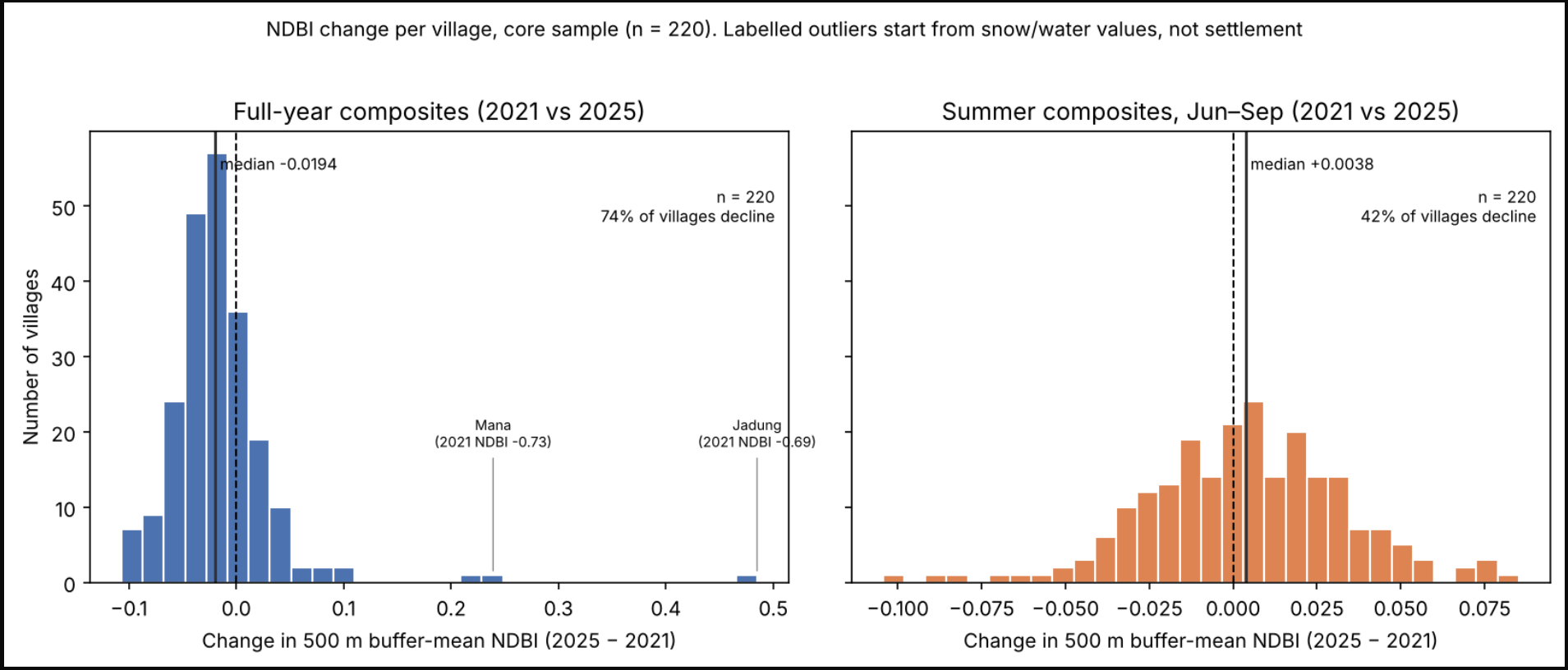
The five Map pages are static renders for print/offline reference. Live versions on the dashboard are interactive (pan, zoom, click-to-inspect).

Figure 1 — Study Area



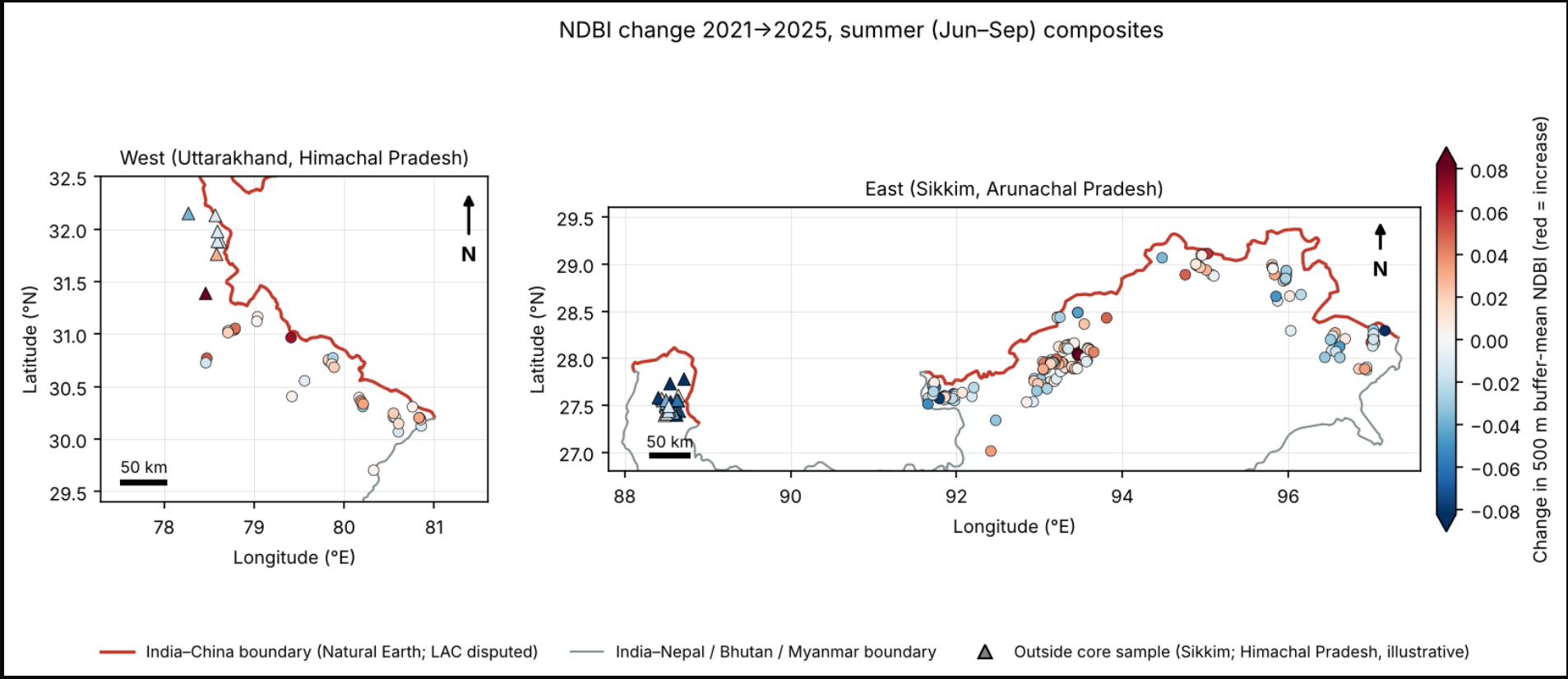
220 core priority villages, 31 Sikkim and 7 Himachal Pradesh villages outside the core sample, and 628 comparison villages.

Figure 2 — NDBI Change per Priority Village



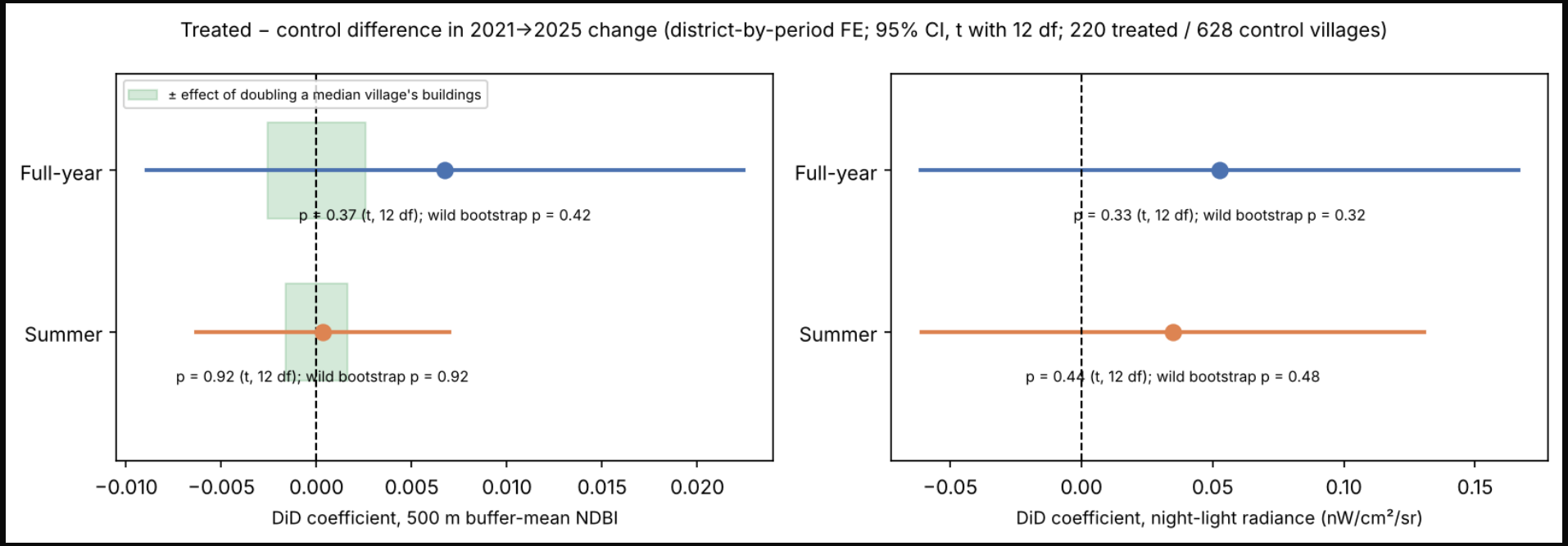
Both compositing windows; labelled outliers start from snow/water values.

Figure 3 — Summer NDBI Change by Village



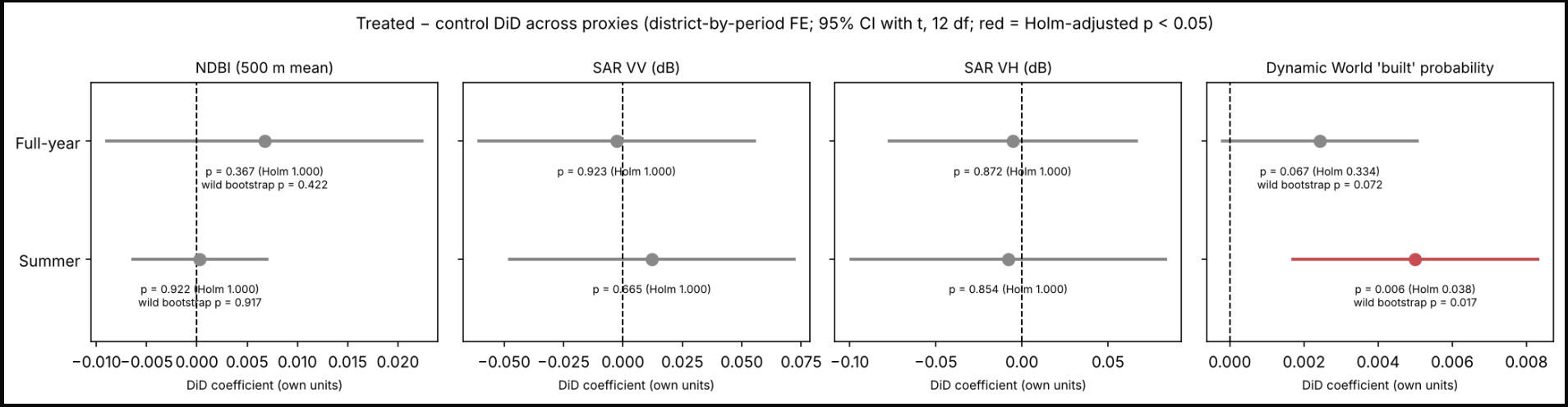
Interactive version: dashboard → Interactive Maps.

Figure 4 — Priority vs Comparison DiD (NDBI, Night Lights)



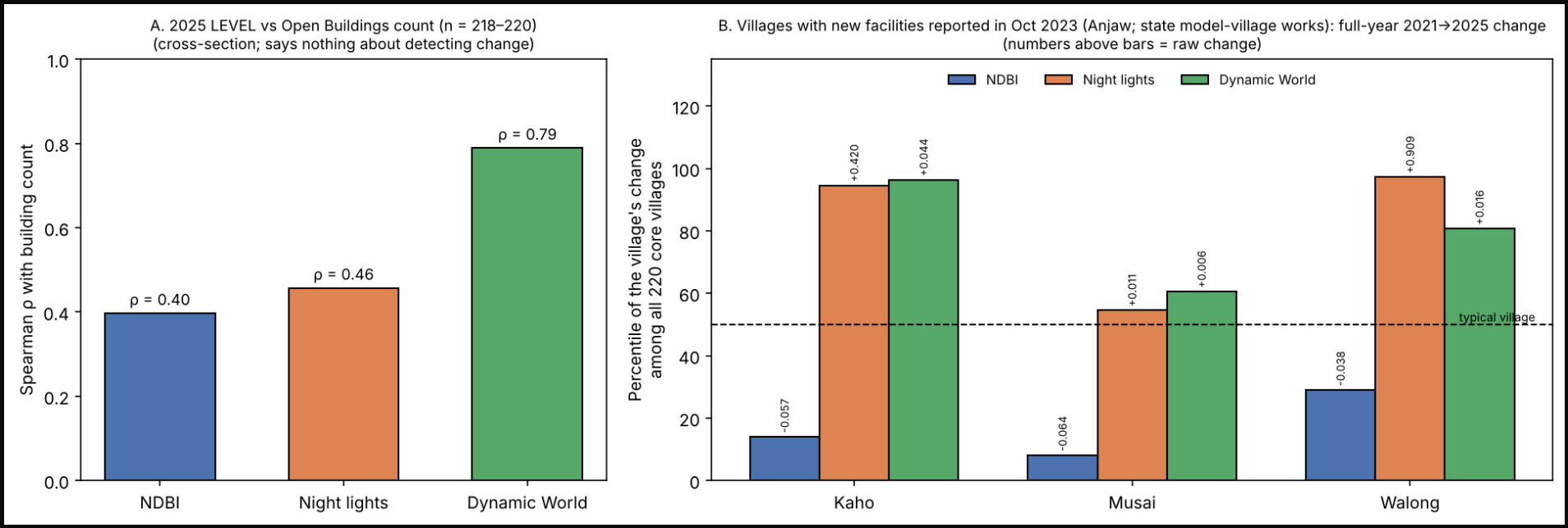
95% CI with t (12 df); green band = change from doubling a typical village's buildings.

Figure 5 — DiD Across Four Satellite Measures



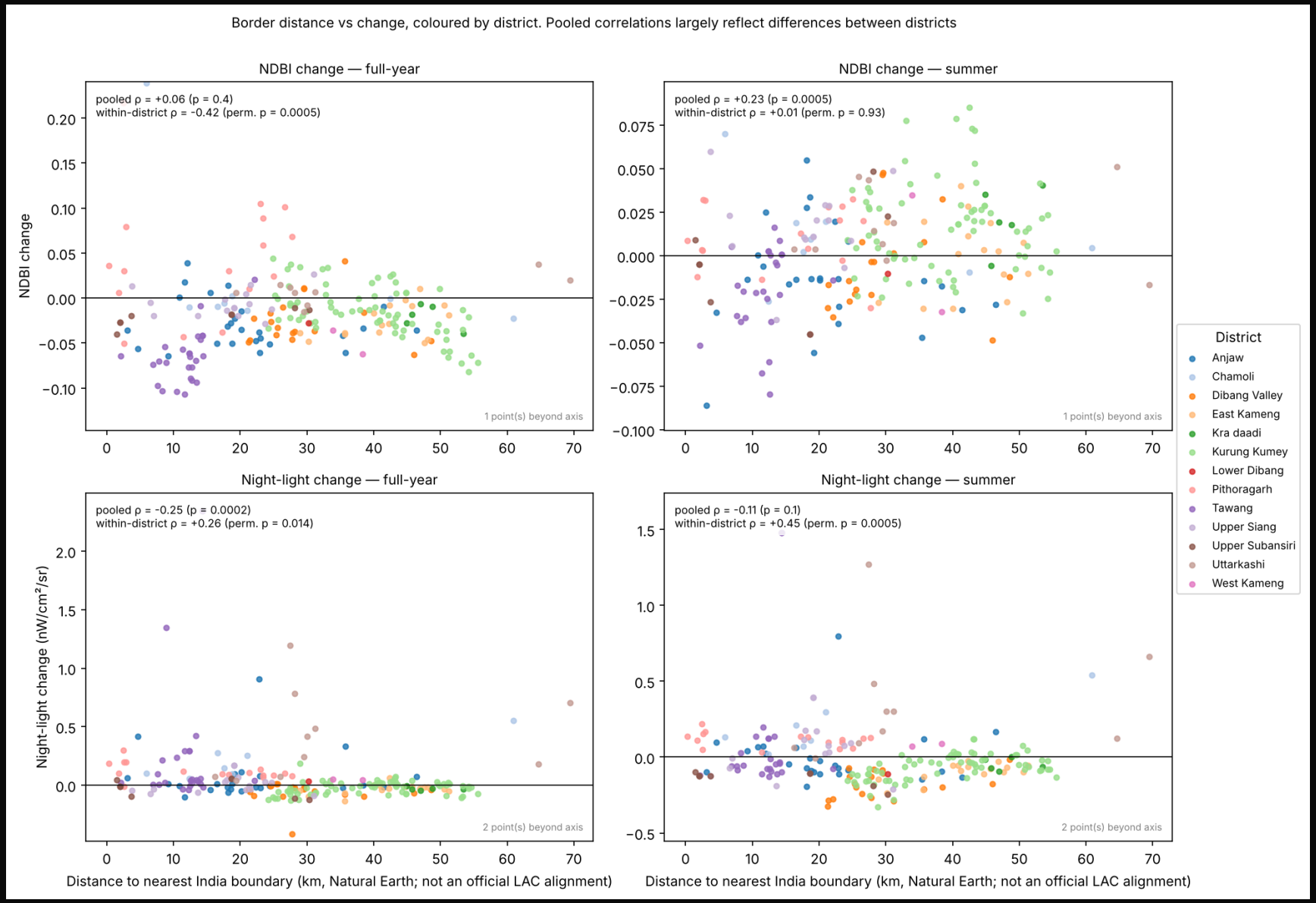
NDBI, Sentinel-1 VV/VH, Dynamic World built probability; red = Holm-adjusted p < 0.05.

Figure 6 — Footprint Correlation and Anjaw Case Villages



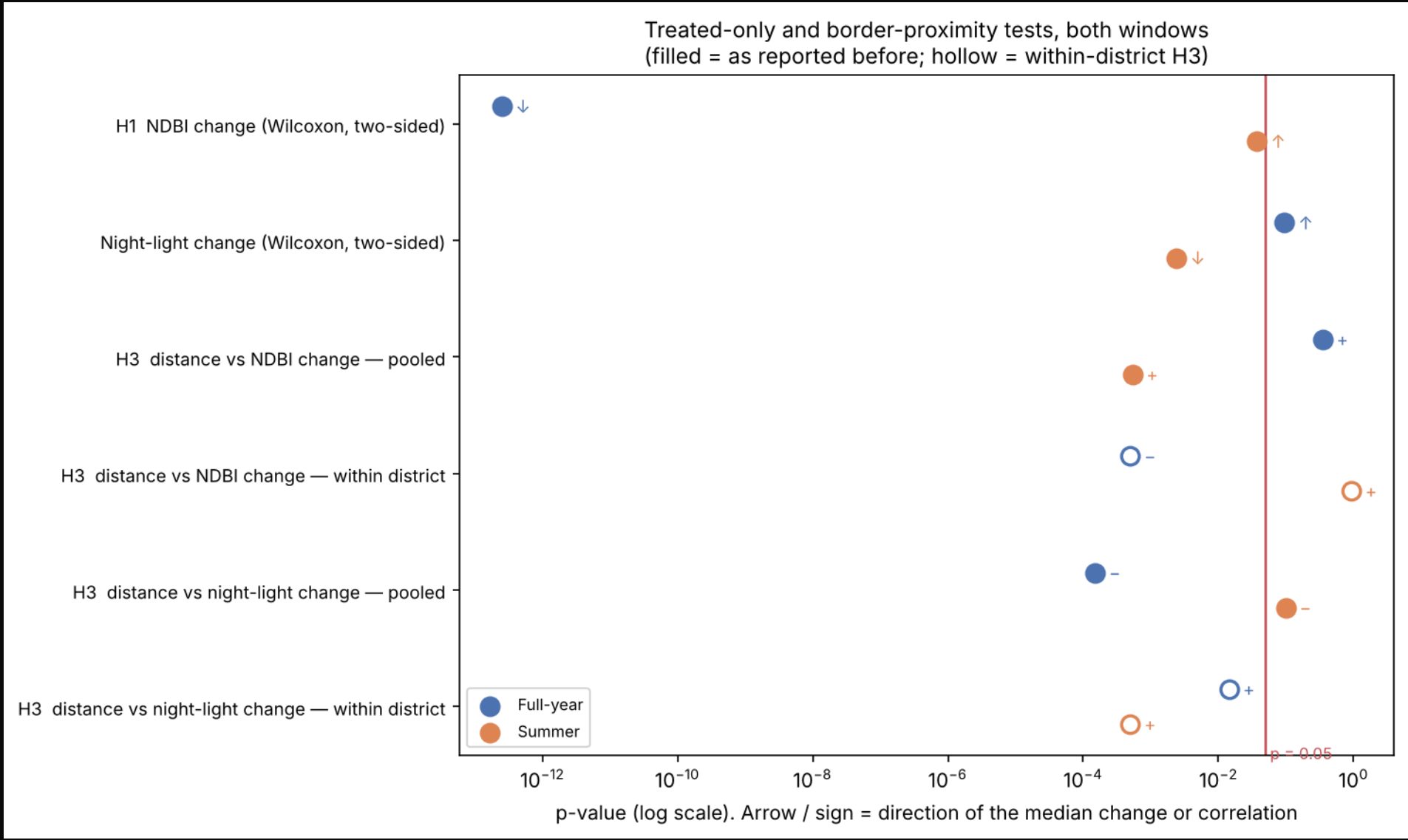
Case villages shown as percentiles of all villages' change.

Figure 7 — Border Distance vs Change, by District



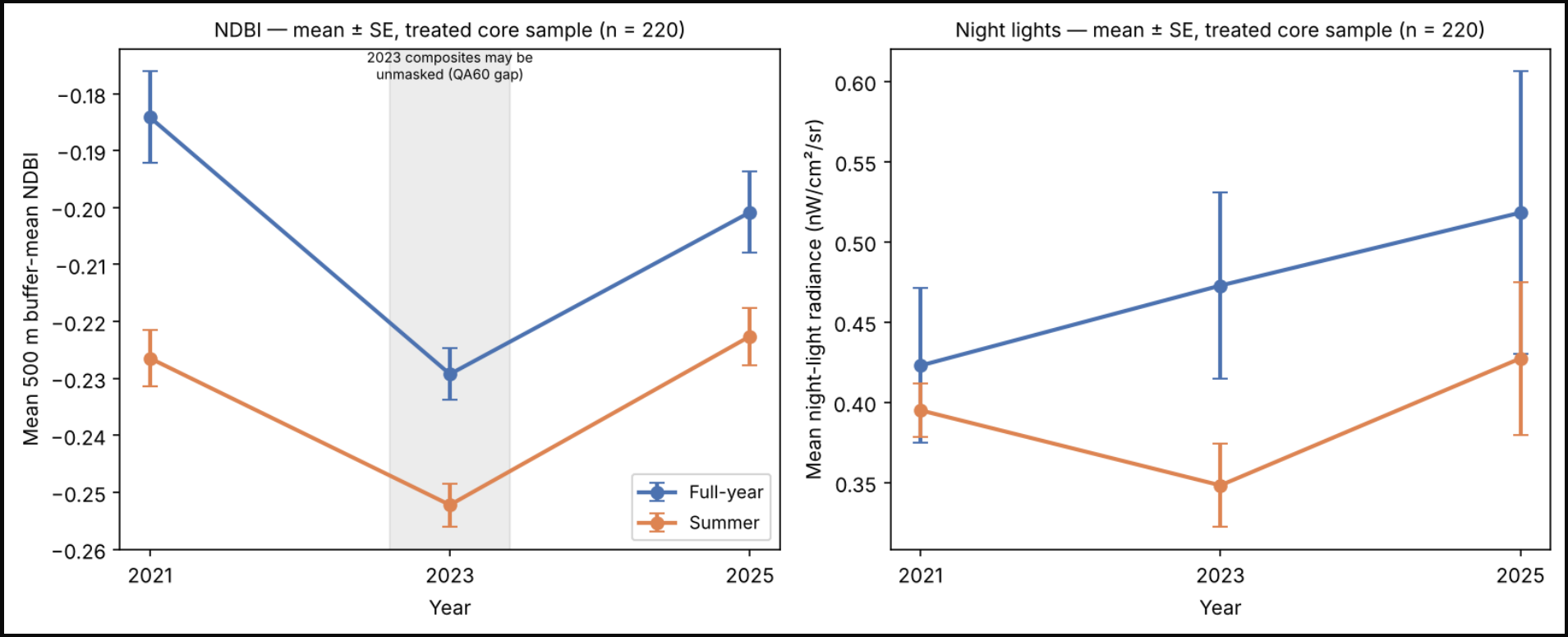
Pooled and within-district Spearman correlations.

Figure 8 — Treated-Only and Border-Distance Tests



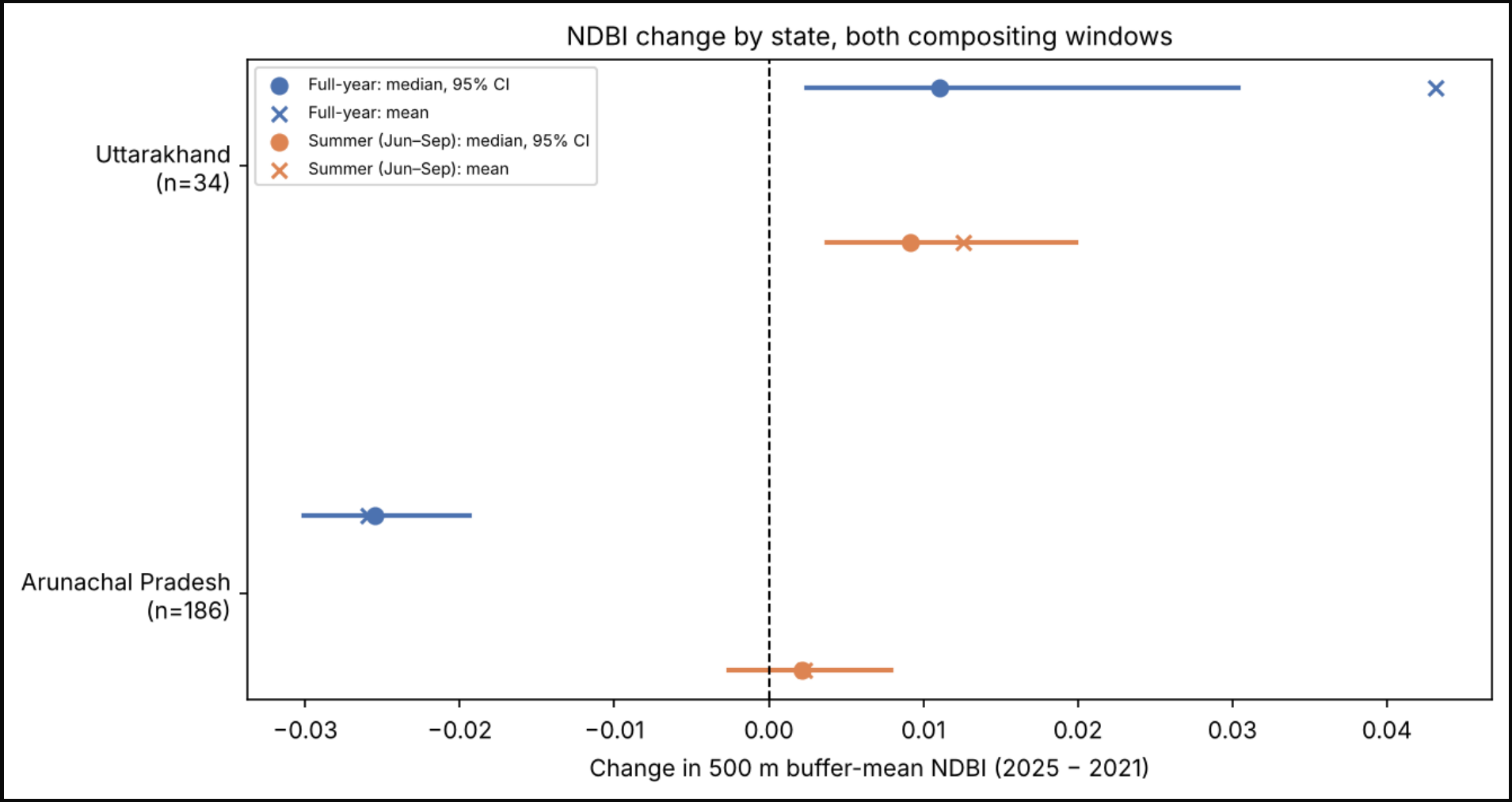
Two-sided tests; within-district H3 shown as hollow markers.

Figure 9 — Three-Point Trend (2021 / 2023 / 2025)



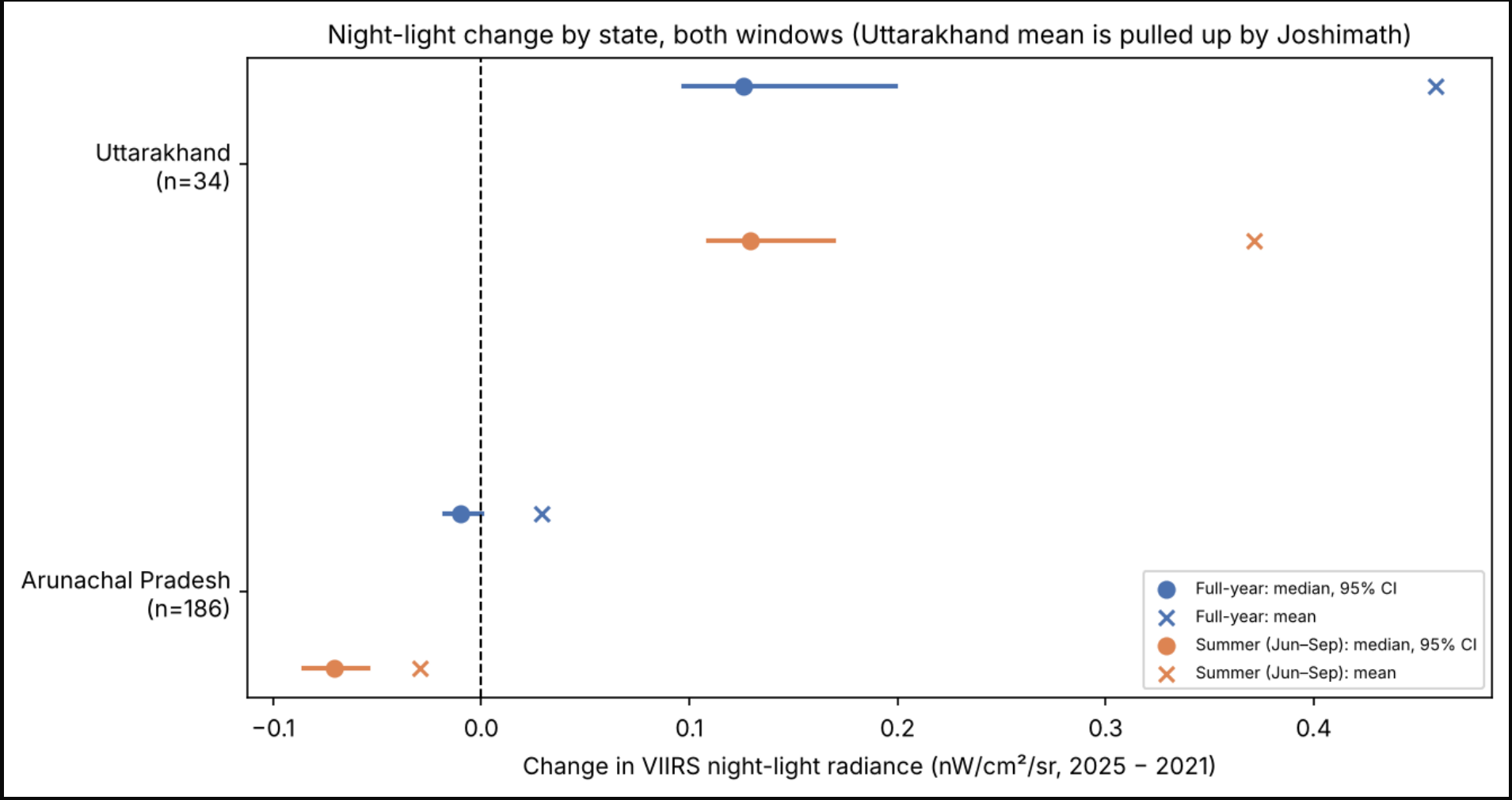
2023 composites may be cloud-unmasked (QA60 gap).

Figure A1 — NDBI Change by State



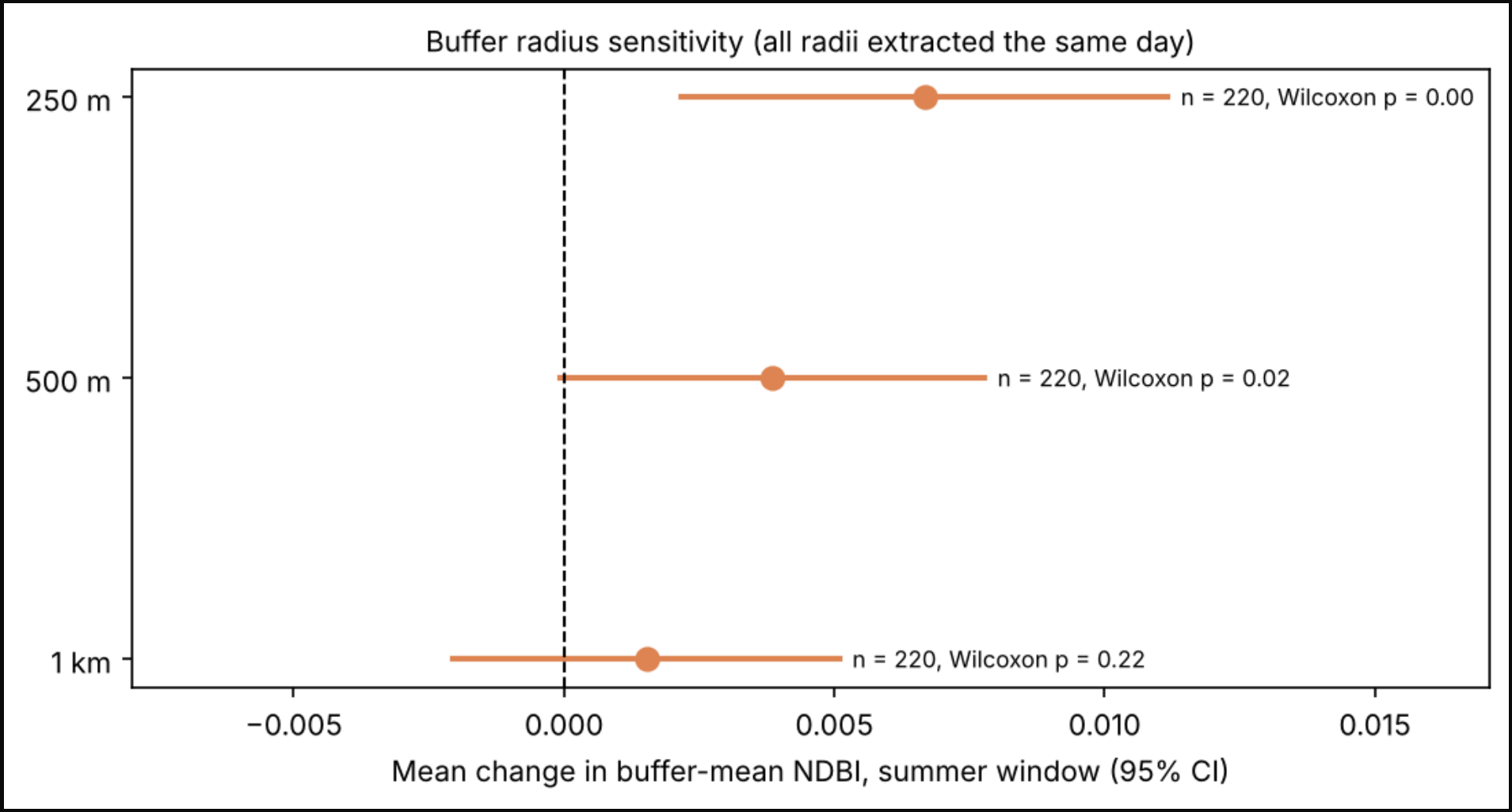
Medians with 95% bootstrap intervals, and means.

Figure A2 — Night-Light Change by State



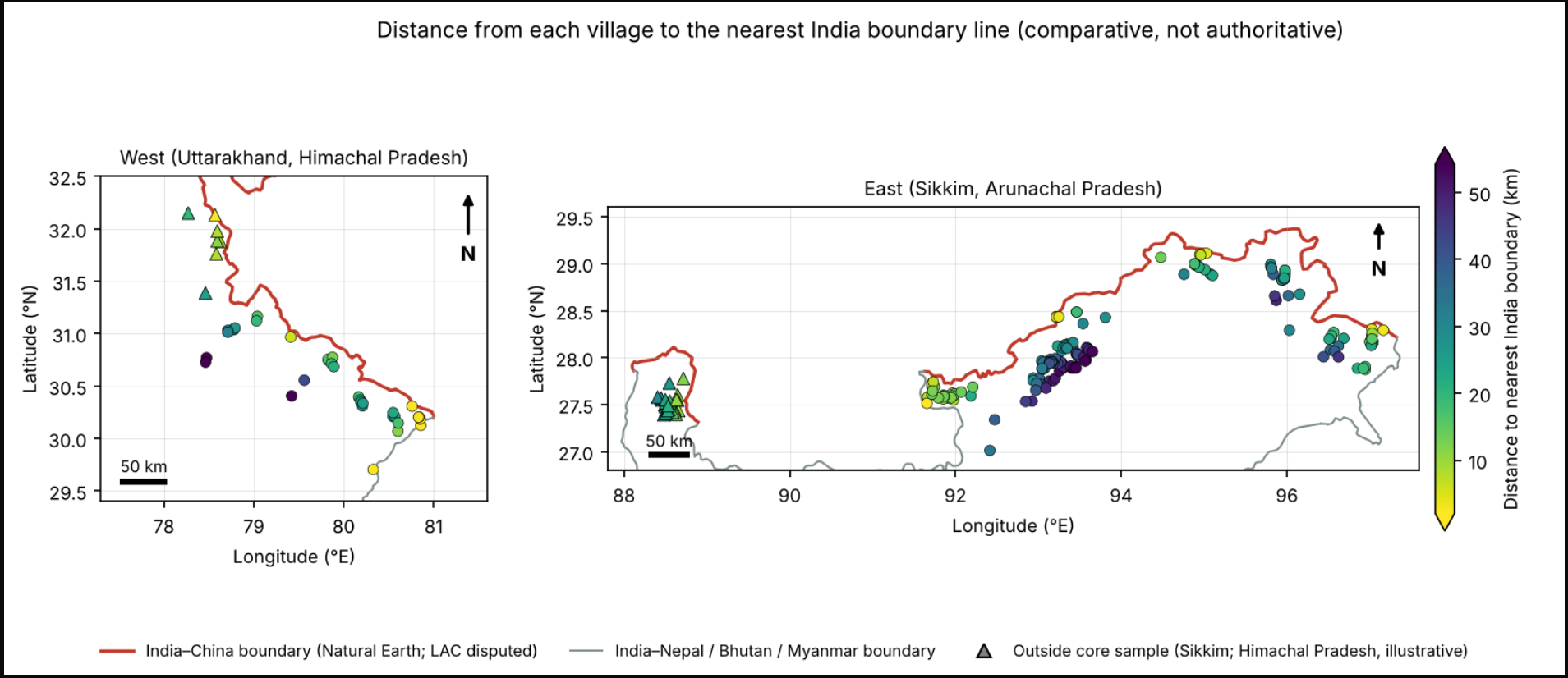
Medians with 95% bootstrap intervals, and means.

Figure A3 — Buffer Radius Sensitivity



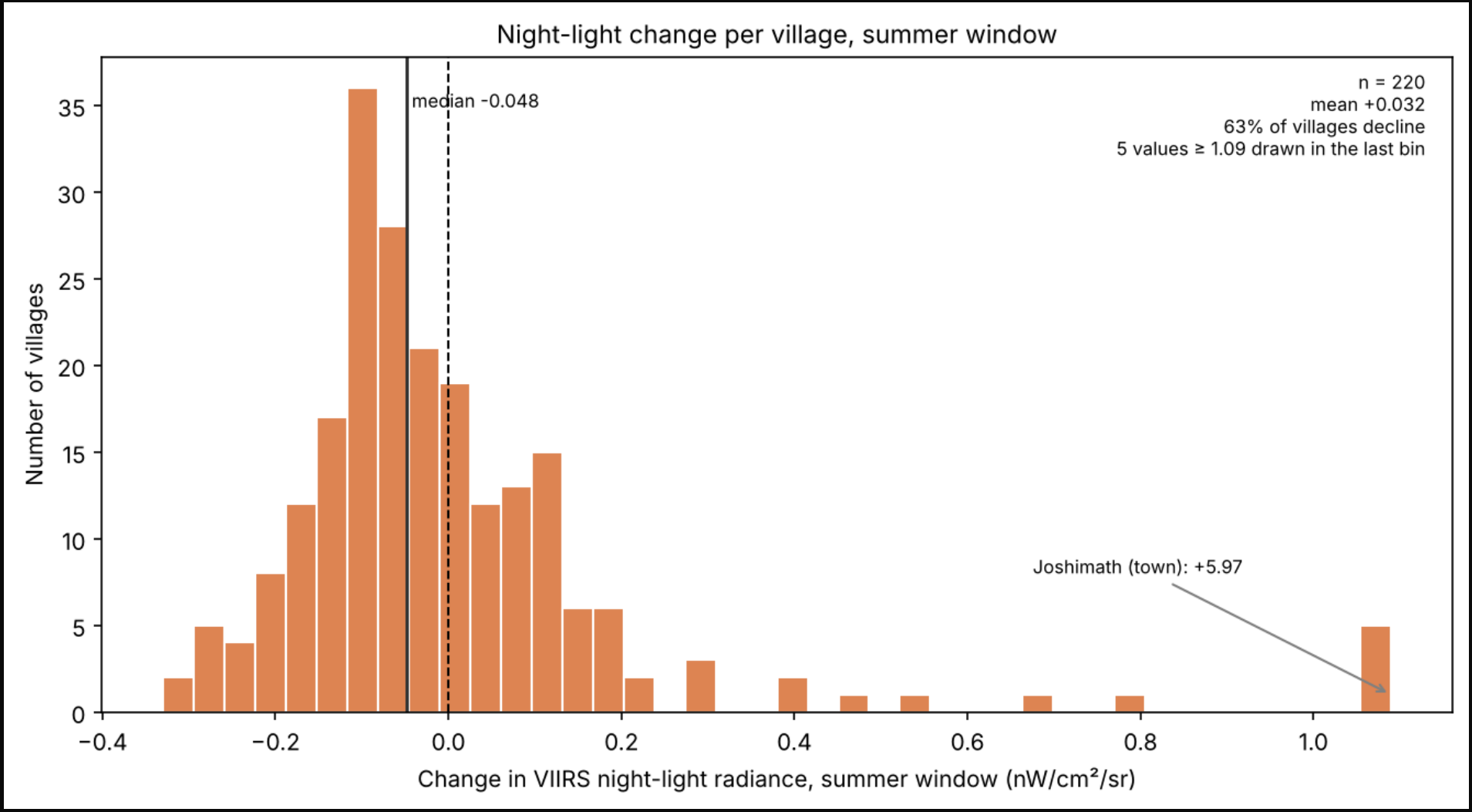
Mean NDBI change with 95% CI at 250 m, 500 m and 1 km (summer).

Figure A4 — Distance to the Nearest Boundary



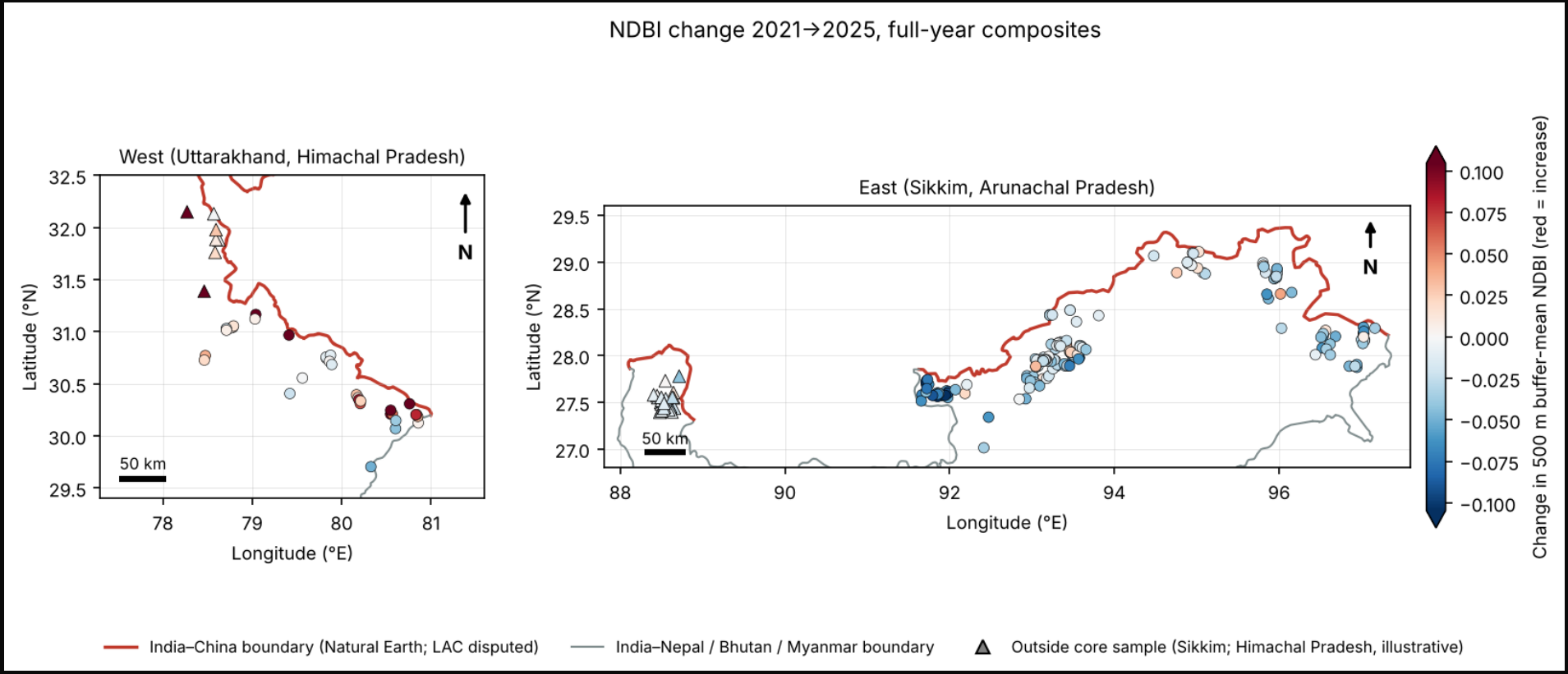
Comparative only; Natural Earth cartographic lines.

Supplementary — Night-Light Change Distribution



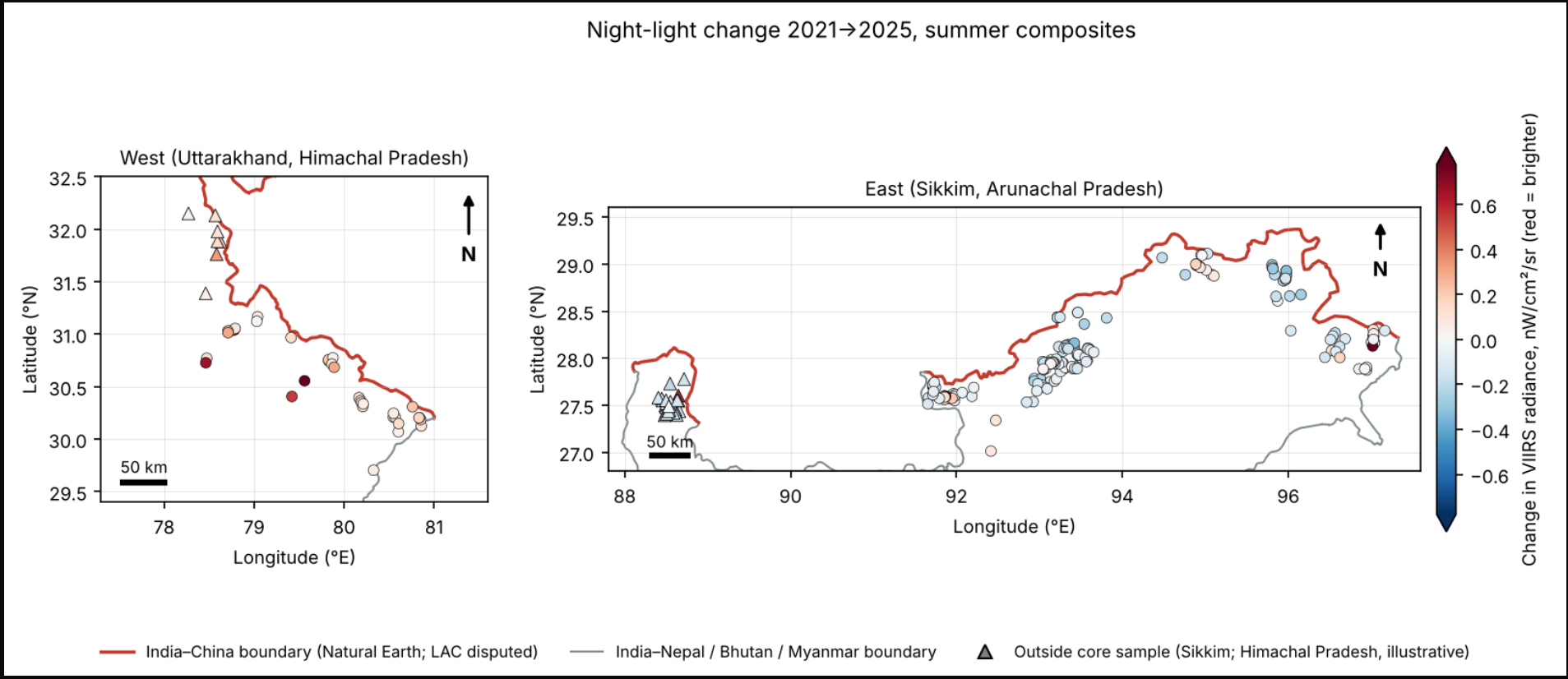
Summer window.

Supplementary — Full-Year NDBI Change by Village



Interactive version: dashboard → Interactive Maps.

Supplementary — Night-Light Change by Village



Interactive version: dashboard → Interactive Maps.